IND-QUASI-AFFINE MORPHISMS COMPOSE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19584-19587 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19589-19603, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Deliberately sequenced after L16109 in the same wave so the cited dictionary card exists on the shelf; short but every step named.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's alternative-definition dictionary for ind-quasi-affine morphisms (card L16109, same wave); Morphisms, lemma composition-quasi-affine.

Source reference. Original lemma 0F1U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19538–19554. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01SN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19603–19603. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The property of being an ind-quasi-affine morphism is stable under composition.*

Proof. Let $f : X \to Y$ and $g : Y \to Z$ be ind-quasi-affine morphisms. Let $V \subset Z$ and $U \subset f^{-1}(g^{-1}(V))$ be quasi-compact opens such that V is also quasi-separated. The image $f(U)$ is a quasi-compact subset of $g^{-1}(V)$, so it is contained in some quasi-compact open $W \subset g^{-1}(V)$ (a union of finitely many affines). We obtain a factorization $U \to W \to V$. The map $W \to V$ is quasi-affine by Lemma 0F1U, so, in particular, W is quasi-separated. Then, by Lemma 0F1U again, $U \to W$ is quasi-affine as well. Consequently, by Morphisms, Lemma 01SN, the composition $U \to V$ is also quasi-affine, and it remains to apply Lemma 0F1U once more. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.